

Week 10 — Activity: From Conservation of Energy to Finite Differences

Geophysics 3A: Potential Fields & Geothermics

May 27, 2026

Purpose

To understand finite difference heat flow equations as direct consequences of energy conservation applied to small volumes of rock.

Consider a vertical column of sediment divided into small layers. Heat flows between layers, and each layer can gain or lose energy depending on its neighbours and any internal heat production.

Rather than solving a continuous equation, we will track how heat moves between *discrete points*.

Part A: Physical intuition

- If a layer is hotter than the ones above and below it, does it gain or lose heat?
- If heat is being produced inside the layer, what happens over time?

Part B: Setting up the system

- Sketch a one-dimensional grid of nodes labeled T_{i-1} , T_i , and T_{i+1} .
- Indicate the spacing Δz .

Part C: Heat flux

Heat flows according to Fourier's law:

$$q_z = -k \frac{\partial T}{\partial z} \quad (1)$$

Tasks:

- Write an expression for heat flowing *into* node i .
- Write an expression for heat flowing *out of* node i .

Part D: Energy conservation

Energy conservation for a small volume gives:

$$\text{Rate of change} = \text{in} - \text{out} + \text{produced}$$

Task: Combine your expressions to derive:

$$\frac{\partial T_i}{\partial t} = \kappa \frac{T_{i+1} - 2T_i + T_{i-1}}{\Delta z^2} + \frac{A_i}{\rho c_p} \quad (2)$$

Part E: Interpretation

Explain in words:

- What does the term $T_{i+1} - 2T_i + T_{i-1}$ represent?
- What role does heat production A_i play?
- What happens if $A_i = 0$?

Key Insight

Finite difference equations are not numerical tricks—they are simply energy conservation applied locally.